PSET 2: Limits and Derivatives

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Limits: left, right, and at a point

Given that[1]

$$\lim_{x \to a} f(x) = -3, \quad \lim_{x \to a} g(x) = 0, \quad \lim_{x \to a} h(x) = 8$$

find the limits that exist. If a limit does not exist, explain why.

a. $\lim_{x \to a} [f(x) + h(x)]$

b. $\lim_{x \to a} \dfrac{g(x)}{f(x)}$

c. $\lim_{x \to a} \dfrac{f(x)}{g(x)}$

d. $\lim_{x \to a} \dfrac{2f(x)}{h(x) - f(x)}$

Problem 2: Find even more limits

Find the following limits:[2]

a. $\lim_{x \to -4} \dfrac{x^2 + 5x + 4}{x^2 + 3x - 4}$

b. $\lim_{x \to 4^-} \sqrt{16 - x^2}$

[1] Grimmer 2012 HW 2.4

[2] Grimmer 2012 HW 2.4(b)

Problem 3: Check for discontinuities

Which of the following functions are continuous? If a function is not continuous, where are the discontinuities?[3]

a. $f(x) = \dfrac{9x^3 - x}{(x-1)(x+1)}$

b. $f(x) = e^{-x^2}$

c. $f(x) = \begin{cases} x^3 + 1, & x > 0 \\ -x^2, & x \leq 0 \end{cases}$

Problem 4: Find finite limits

Find the following finite limits:[4]

a. $\lim_{x \to 1} \left[\dfrac{x^4 - 1}{x - 1} \right]$

Problem 5: Find infinite limits

Find the following infinite limits:[5]

> Hint: use **L'Hôpital's rule** to switch from $\lim_{x\to\infty} \left(\dfrac{f(x)}{g(x)} \right)$ to $\lim_{x\to\infty} \left(\dfrac{f'(x)}{g'(x)} \right)$.

a. $\lim_{x\to\infty} \left[\dfrac{2^x - 3}{2^x + 1} \right]$

b. $\lim_{x\to\infty} \left[\dfrac{3^x}{x^3} \right]$

Problem 6: Assess continuity and differentiability

For each of the following functions, state whether it is continuous and/or differentiable at the point where its two formulas meet.[6]

a. $f(x) = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$

b. $f(x) = \begin{cases} x^3, & x \leq 1 \\ x, & x > 1 \end{cases}$

[3] Gill 1.9
[4] Gill 5.1
[5] Gill 5.3 and 5.8
[6] Simon and Blume 2.16

Problem 7: Calculate derivatives

Recall the differentiation rules:

$$
\begin{aligned}
[kf(x)]' &= kf'(x) && && \text{constant multiple} \\
[f(x) \pm g(x)]' &= f'(x) \pm g'(x) && && \text{sum} \\
[f(x)g(x)]' &= f'(x)g(x) + f(x)g'(x) && && \text{product} \\
\left[\frac{f(x)}{g(x)}\right]' &= \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, && g(x) \neq 0 && \text{quotient} \\
\left[x^k\right]' &= kx^{k-1} && && \text{power}
\end{aligned}
$$

Differentiate the following functions:[7]

a. $f(x) = 4x^3 + 2x^2 + 5x + 11$

b. $y = \sqrt{30}$

c. $h(t) = \log(9t + 1)$

d. $f(x) = \log\left(x^2 e^x\right)$

e. $h(y) = \left(\frac{1}{y^2} - \frac{3}{y^4}\right)\left(y + 5y^3\right)$

f. $g(t) = \frac{3t - 1}{2t + 1}$

Problem 8: Use the product and quotient rules

Differentiate the following function twice—once using the product rule, and once using the quotient rule:[8]

$$f(x) = \frac{x^2 - 2x}{x^4 + 6}$$

Problem 9: Logarithms and exponential functions

Compute the derivative of each of the following functions:[9]

a. $f(x) = \frac{x}{e^x}$

b. $h(x) = \frac{x}{\log(x)}$

[7] Grimmer HW2.3
[8] Grimmer HW2.4
[9] Simon and Blume 5.8

Problem 10: Composite functions

For each of the following pairs of functions $g(x)$ and $h(z)$, write out the composite functions $g(h[z])$ and $h(g[x])$. In each case, describe the domain of the composite function.[10]

a. $g(x) = x^2 + 4, \quad h(z) = 5z - 1$

b. $g(x) = x^3, \quad h(z) = (z-1)(z+1)$

Problem 11: Chain rule

Use the chain rule to compute the derivative of the composite functions in the previous problem from the derivatives of the two component functions. Then compute each derivative directly, using your expression for the composite function. Simplify and compare your answers.[11]

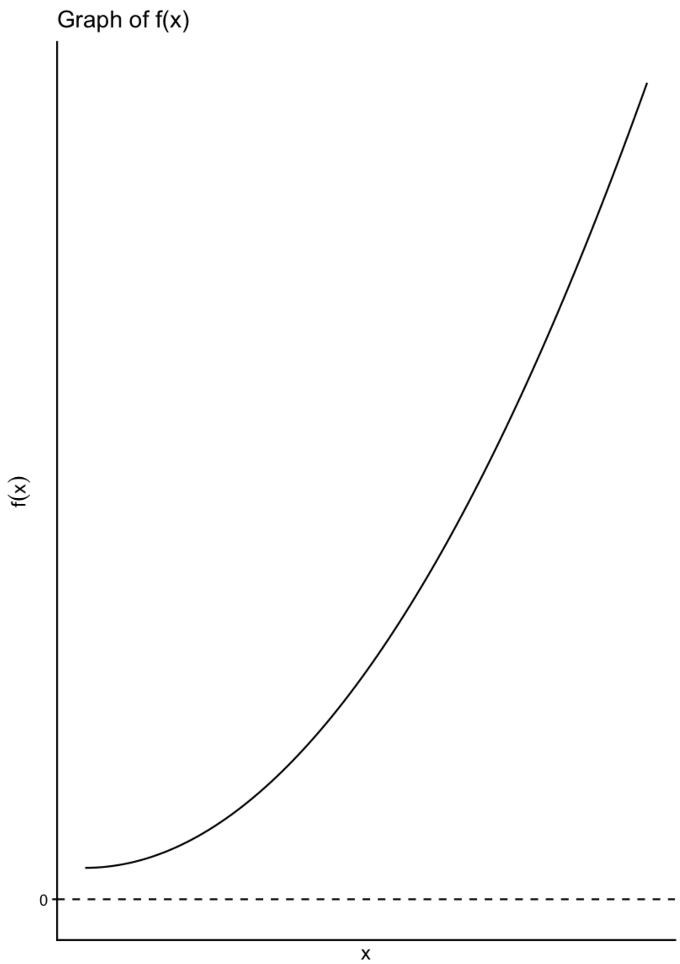
a. $g(x) = x^2 + 4, \quad h(z) = 5z - 1$

b. $g(x) = x^3, \quad h(z) = (z-1)(z+1)$

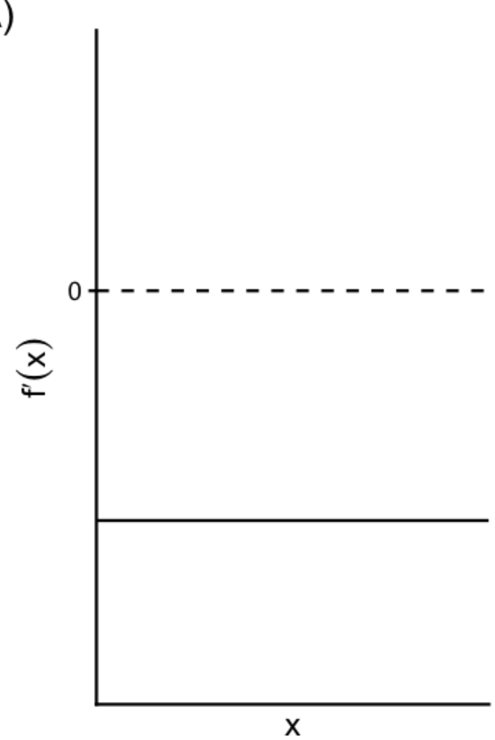
Problem 12: Identify a possible derivative

A friend shows you the graph of a function $f(x)$ below.[12] Which of the following could be a graph of $f'(x)$? For each graph, explain why it might or might not be the derivative of $f(x)$.

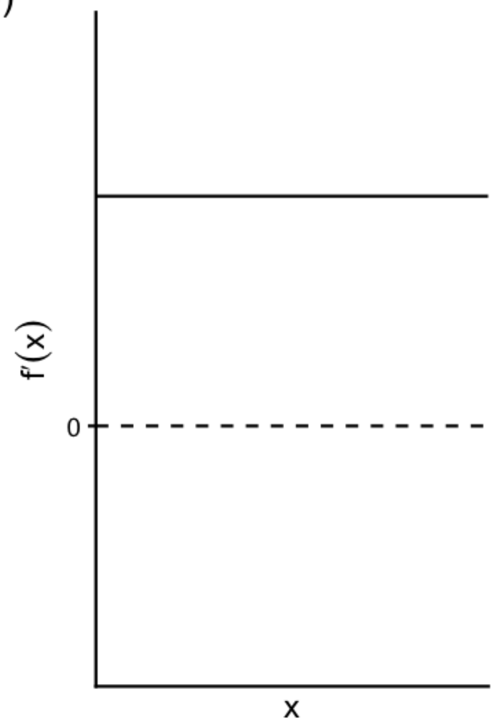
[10] Simon and Blume 4.1
[11] Simon and Blume 4.3
[12] Grimmer HW3.6


Graph of f(x)
f(x)
0
x

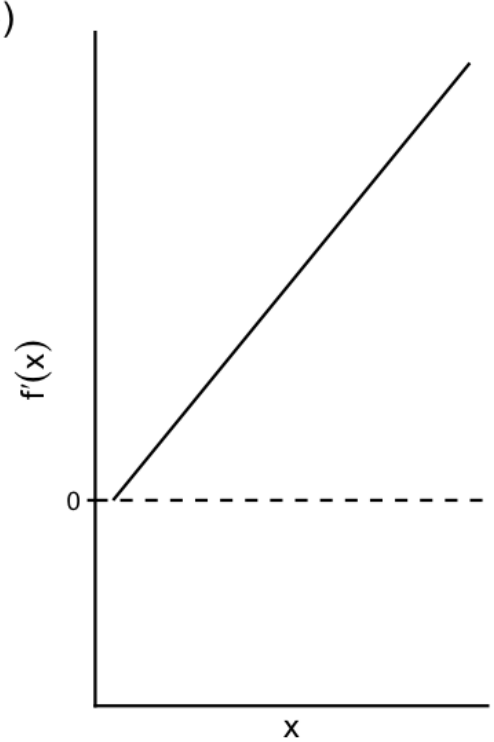
A)



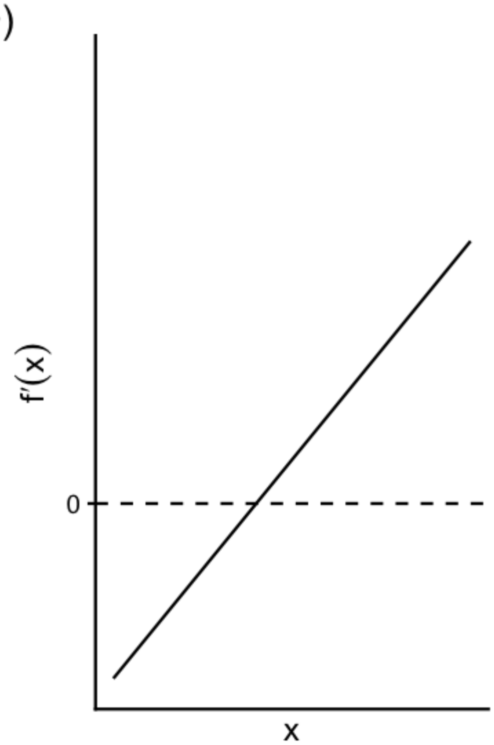
B)



C)



D)



AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.